AYUSH DWIVEDI

B.Tech Undergraduate | Full-Stack & AI Agent Developer

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GitHub: github.com/ayushdwivedi-stack | LinkedIn: linkedin.com/in/ayush-dwivedi | LeetCode (300+)

PROFESSIONAL SUMMARY

Results-driven B.Tech undergraduate at Galgotias College with strong hands-on expertise in Node.js backend development, autonomous AI agent workflows (n8n, MCP, LLM tool-calling), and high-performance algorithms (C++, 300+ LeetCode). Smart India Hackathon (SIH 2025) and HackIndia Spark 4 Finalist with proven ability to design and deploy scalable full-stack applications and automated microservices. Seeking a Software Engineering / Full-Stack Developer Internship to drive high-impact engineering initiatives.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, C++, Python, SQL
- **Backend & APIs:** Node.js, Express.js, RESTful APIs, WebSockets, JWT Authentication, Microservices Architecture
- **AI & Automation:** n8n Workflow Automation, Autonomous AI Agents, Tool / Function Calling, MCP (Model Context Protocol), OpenAI / Gemini API, Prompt Engineering
- **Databases & Storage:** MongoDB (Mongoose), PostgreSQL, Redis (Caching), Supabase
- **Frontend & Web:** React.js, Tailwind CSS, HTML5, CSS3, Responsive UI Design
- **Tools & DevOps:** Git, GitHub Actions, Docker, Linux/Bash, Postman, CI/CD, VS Code
- **Core Concepts:** Data Structures & Algorithms (DSA in C++), OOP, System Design Fundamentals

FEATURED PROJECTS

Autonomous AI Agent & Workflow Automation System | *n8n · Node.js · LLMs · Webhooks · REST APIs*

- Architected an autonomous multi-step AI agent automation pipeline using n8n and Node.js microservices to process events, orchestrate external APIs (GitHub, Gmail, Slack), and execute autonomous decision tasks.
- Implemented intelligent tool-calling and context management with LLM APIs, automating multi-service notification and resolution workflows with 85% time reduction.
- Built resilient webhook listeners, exponential backoff retries, and automated exception logging, ensuring 99.9% uptime for automated tasks.

Full-Stack Enterprise Task & Collaboration Platform | *Node.js · Express · MongoDB · React · Redis · WebSockets*

- Engineered a scalable full-stack web application featuring secure JWT authentication, role-based access control (RBAC), and low-latency RESTful API services in Node.js.
- Integrated Redis caching and MongoDB query indexing, achieving sub-60ms response latencies under concurrent request testing.
- Delivered a real-time reactive frontend using React.js and Tailwind CSS with WebSocket channels for collaborative project management.

Model Context Protocol (MCP) Integration Engine | *Node.js · Python · Google Workspace API · GitHub API · stdio*

- Engineered and deployed a custom Model Context Protocol (MCP) server that empowers AI assistants to execute authenticated workspace operations directly via standardized stdio channels.
- Integrated robust OAuth 2.0 flow, token caching, secure MIME parsing, and structured JSON-RPC responses for seamless tool invocation.

AI-Powered Wafer Quality Inspection System | *Node.js · Python · Computer Vision · Deep Learning · REST APIs*

- Developed a computer-vision inspection pipeline to automate semiconductor wafer defect classification, utilizing deep learning models paired with a high-throughput Node.js/Express API service.
- Designed an interactive dashboard with live defect category scoring and analytics visualization for manufacturing QA tracking.

EDUCATION

Galgotias College of Engineering & Technology, Greater Noida

Bachelor of Technology (B.Tech) | CGPA: 8.0 / 10.0 | 2024 – 2028

ACHIEVEMENTS & COMPETITIVE HIGHLIGHTS

- Smart India Hackathon 2025 (SIH 2025): Selected to represent college in India's flagship national hackathon, building scalable tech under a 36-hour sprint.
- Finalist — HackIndia Spark 4: Placed among top teams in multi-college hackathon, recognized for rapid MVP engineering and prototype deployment.

• Competitive Programming: Solved 300+ problems on LeetCode & Codeforces with strong foundation in Graphs, Dynamic Programming, and C++ optimization.

• Open-Source Contributor: Active GitHub contributor across developer automation repositories and modern developer tooling.